Roman Penunuri

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Portfolio: rap1018-sudo.github.io/roman-penunuri.github.io

SUMMARY

Electrical Engineering student at the University of Houston (Power & Energy Systems, GPA 3.82) with hands-on analog and embedded circuit design experience, including a BJT transistor logic circuit built and debugged from first principles, plus industrial-scale relay and switchgear work at Air Liquide. Comfortable across the stack from Falstad simulation and breadboard prototyping to embedded C on ARM Cortex-M4. OSHA-30 certified. Interested in circuit design, electronics, and technical R&D roles.

EDUCATION

University of Houston, Cullen College of Engineering — Houston, TX

B.S./M.S. Electrical Engineering (Accelerated Program), Power & Energy Systems Concentration · GPA: 3.82 · Expected Graduation: December 2028

Dean's List: Fall 2024, Spring 2026

WORK EXPERIENCE

Tesla — Electrical Infrastructure Engineer Intern

Gigafactory Texas — Austin, TX · August 2026 – December 2026 (Upcoming)

- Supporting electrical infrastructure design from the utility service entrance down to end-user load, including load calculations, cable/conductor sizing, single-line diagrams, panel schedules, and facility layouts in AutoCAD and Revit, plus verifying designs via field surveys and developing as-built documentation across multiple concurrent projects.

Air Liquide — Technical Support Electrical Engineering Intern, Large Industries Group

Houston, TX · June 2026 – August 2026

- Reprogrammed setpoints on 9 protective relays (8 microprocessor, 1 electromechanical) protecting generators, crank motors, transformers, and feeder breakers using GE Vernova EnerVista, basing settings on comparable units and resetting unused parameters to reduce configuration confusion.
- Redlined one-line diagrams, three-line diagrams, control schematics, breaker schematics, and panel door layouts to define contractor scope for a relay upgrade program, and updated relay/transformer settings in an SKM Power*Tools model to reflect as-found and proposed states for arc flash hazard analysis and PPE labeling, pending PE verification.
- Conducted electrical fire risk assessments at 4 plants against NFPA standards, documenting transformer containment and electrical room fire-safety compliance via field checklists.
- Built a custom Gemini Gem — an AI auditor with anti-hallucination guardrails and citation requirements — to audit 24 electrical safety training modules against internal documentation and NFPA 70E/CSA Z462 standards, cutting manual audit time from weeks to about a week; authored best-practice documentation for the team.

Air Liquide — Plant Operations Electrical Engineering Intern, Bayport Plant

Pasadena, TX · June 2025 – August 2025

- Identified failure modes in motor protection systems and implemented upgraded relay settings via GE Vernova (Multilin/EnerVista), eliminating a recurring equipment defect and reducing projected maintenance costs by ~$10,000.
- Authored scope-of-work documentation for plant turnaround operations covering motors, switchgear, and battery systems in coordination with Engineering, Safety, and Maintenance teams; mitigated potential operational losses exceeding $1M.
- Revised NFPA 70B and IEEE-compliant maintenance standards to shift asset management from time-based to condition-based scheduling, reducing unnecessary testing cycles and extending equipment lifecycle; selected for return internship based on performance.

PROJECTS

Water Level Indicator

ECE 3155 Electronics Lab — University of Houston

Designed a BJT transistor logic circuit activating LEDs at discrete water levels for a 5-person team project; modeled node voltages in the Falstad Circuit Simulator before bench testing with oscilloscopes and multimeters, then diagnosed dim LED output by probing nodes against the model, isolating the root cause to distilled water resistivity without replacing components.

PANTRi — Household Inventory App

Conceived and used AI-assisted development tools to build and ship a full-stack household inventory app; tracks real-time inventory, auto-generates shopping lists, and surfaces spend/waste analytics to cut unnecessary purchases. Released to the App Store, with Google Play in testing.

SKILLS

Circuits: Analog Circuit Design (BJT Logic), Circuit Simulation (Falstad), Schematic Design, Breadboard Prototyping, Oscilloscope & Lab Equipment

Embedded Systems: ARM Cortex-M4, ARM Assembly, Embedded C, Microcontroller Programming, State Machine Design, Numerical Methods

Certifications: OSHA-30 Hour · PLC Programming (Automation Community) · Pursuing: FE Exam / PE License

Other: MATLAB, GE Vernova (Multilin/EnerVista), SKM Power*Tools, Git & GitHub

ORGANIZATIONS

Society of Hispanic Professional Engineers (SHPE) — August 2023 – Present | Tau Beta Pi Honor Society — March 2025 – Present | IEEE UH Chapter — October 2025 – Present